Alexandra Schuller, CPI

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SKILLS AND QUALIFICATIONS

Languages Bilingual in French and English, with native-level proficiency in both.
Modelling and Simulation | ETABS • SAFE • SAP2000 • CSI Bridge • WoodWorks • Revit • AutoCAD • Abaqus • DIANA • SUMO • HEC-RAS
Standards | NBCC 2020 (Part 4) • CSA A23.3 • CSA S16 • CSA O86 Engineering Design in Wood • CSA S6 • CSA S7:23
Programming | Python • MATLAB • VBA (Excel) • Octave • GitHub • Claude Code • WordPress
Project Management | MS Project • Primavera P6 • Procore • Aconex • Bluebeam • Stack • Asana • Notion
Other | Accpac/Sage • Concur • Windows • Linux • MacOS
Licences & Certifications | Driver's licence • ASP Construction Health and Safety (2022) • ASP Lockout • WHMIS for laboratory personnel (Concordia)

EDUCATION

Bachelor of Engineering, Civil Engineering (Structures), with Distinction **2022-2026**
Co-operative Education Program (Co-op)
Concordia University, Montreal, QC
Member of the Institute for Co-operative Education

University Prerequisite Program for Adults, Continuing Education **2021-2022**
Dawson College, Semester Honours

Bachelor of Arts, Classical Studies, Specialization in Archaeology **2009-2013**
McGill University, Montreal, QC

DEC in Arts and Sciences **2007-2009**
Marianopolis College, Montreal, QC

DESIGN PROJECTS

10-Storey Reinforced Concrete Office Building, Quebec City **September - December 2025**
Course project, Design of Reinforced Concrete Structures (CIVI 453), Concordia University
Complete structural design of a 31.25 m × 33.75 m office building (flat slabs, columns and shear walls) to NBCC 2020 and CSA A23.3.

- Calculated dead, live, snow, wind and seismic loads (including torsional effects) and sized the interior, edge and corner columns.
- Designed a typical slab using the Direct Design Method and validated it with finite element analysis (SAFE): positive and negative reinforcement, punching shear and reinforcement drawings.
- Built a 3D model in ETABS, compared base shear (equivalent static method and response spectrum analysis) and designed a ductile shear wall with reinforcement details.

CapDécarie, Civil Engineering Capstone Project **September 2025 - April 2026**
Took the initiative to propose a project outside the options offered, then recruited and rallied a team of five around the concept: covering a section of the Décarie Expressway trench, between Snowdon and Queen Mary, with a 5,670 m^2 pedestrian park, and replacing the aging Queen Mary overpass.

- Responsible for the design of the foundations and beams of the cap structure, the drainage design and the traffic simulation for construction staging.
- Validated structural behaviour with finite element modelling (CSI Bridge, SAP2000); steel and reinforced concrete design, load analysis and seismic analysis.
- Also covered embodied carbon, cost estimating and construction planning.

Captain, Concordia Concrete Canoe Team **September 2023 - May 2026**

Revived the team after seven years of inactivity and led twelve people in designing, building and transporting a 6-metre concrete canoe to the national competition in Quebec City.

- Concrete mix design, laboratory testing and structural analysis of the canoe.
- Managed the budget, material procurement and lab safety despite tight constraints on space, tools and funding.

WORK EXPERIENCE

Intern, Direction générale des barrages **April - August 2025**
Major Projects Directorate
Ministry of the Environment, the Fight Against Climate Change, Wildlife and Parks (Quebec)

Helped implement a new centralized MS Project planning system for about 1,000 dams in Quebec by consolidating project schedules and milestones with Python and AI tools.

Intern, Renouveau Lafontaine - Louis-Hippolyte-La Fontaine Tunnel **January - April 2025**
Pomerleau

Tracked the progress of heating cable repairs in the main and access tunnels; built a Python tree of the tunnel's electrical systems that automatically generates their power supply diagrams.

Intern, Local Energy Efficiency Partnerships (LEEP) **September - December 2023**
Natural Resources Canada
CanmetENERGY, Ottawa

Designed a VBA data analysis tool (costs, GHG, energy) for converting homes to heat pumps, which uncovered an error in published government data and led to a revision.

Program Assistant - The McConnell Foundation **January 2015 - August 2021**
Montreal, QC

Supported the Cities for People program, which funds projects that improve urban life across Canada.

Intern & Research Assistant - Université de Montréal **Summers 2011 and 2012**

Archaeological excavations at *Argilos*
Surveyed the site, laid out trenches and documented the excavations following archaeological protocols.

AWARDS & DISTINCTIONS

Graduated with Distinction - Bachelor of Engineering, Civil, Concordia University **June 2026**

Pierre Sauvé, Eng. Scholarship for the Advancement of Women in Engineering **March 2023**
Fondation de l'Ordre des ingénieurs du Québec (FOIQ)

Valedictorian - Centennial Regional High School (CRHS) **June 2007**

PROFESSIONAL ASSOCIATIONS

Candidate to the Engineering Profession (CPI) - Ordre des ingénieurs du Québec **September 2026**

Board Member **December 2022 - May 2023**
The Concordia Greenhouse, Montreal, Quebec
Member of the infrastructure committee overseeing options to upgrade the greenhouse's existing components.

OTHER INTERESTS

Sports Hockey, softball and tennis; Triathlon Montréal Esprit Sprint (2020)
Other Passions Greenhouse design, archaeology, community gardens, music (200+ performances)

REFERENCES

Dr. Archie Huang Assistant Professor, Building, Civil, and Environmental Engineering • archie.huang@concordia.ca
Dr. Adel Hanna Professor, Building, Civil, and Environmental Engineering • am.hanna@concordia.ca